

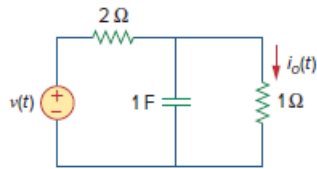
## Problem

If the input voltage in the circuit of Fig. 17.24 is

$$v(t) = \frac{7}{3} + \frac{1}{\pi^2} \sum_{n=1}^{\infty} \left( \frac{1}{n^2} \cos nt - \frac{\pi}{n} \sin nt \right) \text{ V}$$

determine the response  $i_o(t)$ .

**Figure 17.24:**



## Step-by-step solution

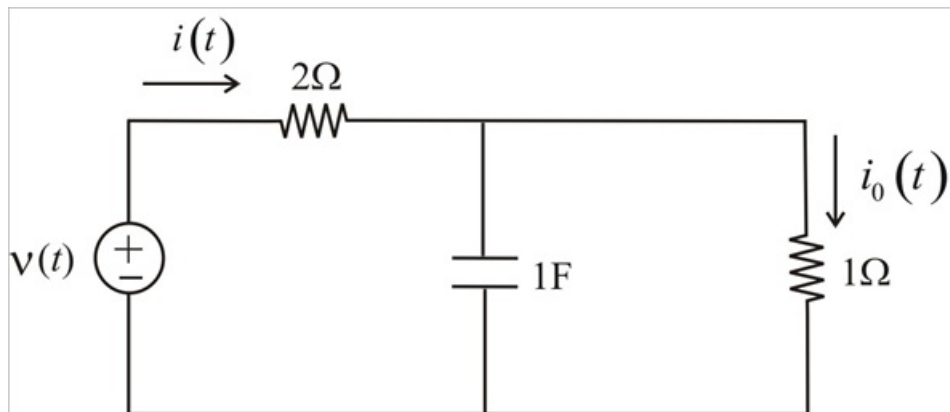
## Step 1 of 6

758-17-7PP SA code: 4477

SR code: 4979

## Step 2 of 6

Given circuit is,



## Step 3 of 6

Given input voltage in the circuit

$$v(t) = \frac{1}{3} + \frac{1}{\pi^2} \sum_{n=1}^{\infty} \left[ \frac{1}{n^2} \cos(nt) - \frac{\pi}{n} \sin(nt) \right] V$$

Where  $\omega_0 = 1$

$$\omega_n = n \text{ rad/sec}$$

The impedance at the source is

$$Z = \left[ 1 \parallel \frac{1}{j\omega_n C} \right] + 2$$

$$Z = \frac{1 \times \frac{1}{j\omega_n C}}{1 + \frac{1}{j\omega_n C}} + 2$$

$$Z = \frac{\frac{1}{jn}}{1 + \frac{1}{jn}} + 2$$

$$Z = \frac{1}{1 + jn} + 2$$

$$Z = \frac{3 + j2n}{1 + jn}$$

#### Step 4 of 6

The input current

$$I = \frac{V}{Z}$$

$$I = \frac{1 + jn}{3 + 2jn} V$$

Where  $V$  is the phasor form of the source voltage  $v(t)$ .

By the current division,

$$I_0 = \frac{\frac{1}{jn}}{1 + \frac{1}{jn}} I$$

$$I_0 = \left[ \frac{1}{1 + jn} \right] \left[ \frac{1 + jn}{3 + 2jn} \right] V$$

$$I_0 = \left( \frac{1}{3 + 2jn} \right) V$$

$I_0$  can be expressed as,

$$I_0 = \frac{V}{\sqrt{9 + 4n^2} \angle \tan^{-1} \left( \frac{2n}{3} \right)}$$

#### Step 5 of 6

For the  $dc$  component ( $\omega_n = 0$  or  $n = 0$ )

$$V = \frac{1}{3}$$

$$I_0 = \frac{V}{3}$$

$$I_0 = \frac{1}{9}$$

$$\therefore I_0 = \frac{1}{9}$$

#### Step 6 of 6

For the  $n^{th}$  harmonic,

$$V = \frac{1}{\pi^2} \sqrt{\frac{1}{n^4} + \frac{\pi^2}{n^2}} \angle -\tan^{-1} \left( \frac{\frac{-\pi}{n}}{\frac{1}{n^2}} \right)$$

$$V = \frac{1}{n^2 \pi^2} \sqrt{1 + n^2 \pi^2} \angle \tan^{-1} (n\pi)$$

$$I_0 = \left\{ \frac{1}{\sqrt{9 + 4n^2} \angle \tan^{-1} \left( \frac{2n}{3} \right)} \right\} \left\{ \frac{\sqrt{1 + n^2 \pi^2} \angle \tan^{-1} (n\pi)}{n^2 \pi^2} \right\}$$

$$I_0 = \frac{\sqrt{1 + n^2 \pi^2}}{n^2 \pi^2 \sqrt{9 + 4n^2}} \angle \tan^{-1} (n\pi) - \tan^{-1} \left( \frac{2n}{3} \right)$$

Therefore, the response  $i_0(t)$  in time domain

$$i_0(t) = \frac{1}{9} + \sum_{n=1}^{\infty} \frac{\sqrt{1 + n^2 \pi^2}}{n^2 \pi^2 \sqrt{9 + 4n^2}} \cos \left( nt - \tan^{-1} \left( \frac{2n}{3} \right) + \tan^{-1} (n\pi) \right) A$$